producing fast reactors (EBR I) went critical in 1951. In the GEN IV system it is assumed that there is a clear integration of both thermal reactors as well as fast reactors, recycling plants for the fuel and fuel fabrication. Regarding the fast reactors some "novel" combinations of fuel and coolant are currently discussed. The proposed recycling plats could also be designed to avoid a pure plutonium steam for proliferation resistance using a separation process similar to the Group Actinide EXtraction (GANEX) process, see Chapter 21. However, with the development of centrifuge based uranium enrichment, the proliferation path using reactor produced plutonium will become less and less attractive for countries desiring nuclear weapons.

20.1. THERMAL REACTORS

The large majority of nuclear power reactors is based on fission by thermal neutrons and the discussion will begin with this class of reactors.

20.1.1. Pressurized water reactors

As mentioned in Chapter 19, the name implies that a pressurized water reactor is cooled by hot high pressure water, either H_2O (PWR, VVER) or D_2O (PHWR). In the PWR and VVER types the coolant is also used as moderator whereas a separate D_2O containing moderator tank is normally used in the PHWR type. These power reactor types have several things in common: primary − secondary coolant circuits separated by heat exchangers (*steam generators*), a *pressurizer* to adjust primary system pressure, and often *chemical shim control* for adjustment of the excess reactivity with fresh fuel.

The pressurized water reactor is generally preferred for propulsion purposes (military surface vessels and submarines), partly because it can react faster to changes in power demand than many other types of thermal reactors.

(a) *PWR*. The main components of a pressurized light water moderated and cooled reactor (PWR) station have been described in §19.1 and Figure 19.2.

The important features of the PWR core are shown in Figure 19.12 and described in §19.13.1. The design with a pressure vessel without facilities for fuel handling under pressure necessitates a yearly shutdown for fuel replacement. However, this is not considered a great inconvenience as a PWR anyhow has to be shutdown on a regular basis for safety inspection of pipes, welds, etc, and for routine maintenance. The need to shut down the reactor and open the pressure vessel in order to replace fuel also makes the PWR (as well as the VVER) resistant to concealed nuclear proliferation.

Leaking tubes in the steam generators have plagued many PWRs. The leaks are usually caused by corrosion. Such leaks are normally fixed by plugging the affected tube. However, too many plugged tubes leads to a reduction in permitted power

output; finally requiring either permanent shut down or replacement of the steam generators. In the beginning of 2000 many PWRs have already had their steam generators replaced (52 reactors) and replacements for another 21 reactors was planned.

Another generic problem has been the development of cracks in the control rod guide tubes of inconel which penetrate a PWR pressure vessel head. Such cracks are difficult to locate and repair because of the confined space between the tubes. After some time leaks usually develop when the cracks grow larger, ultimately requiring replacement of the vessel head. Crack formation can probably be prevented by a proper selection of material, its pre-treatment and meticulous adjustment of the water chemistry.

(b) *VVER*. In the former USSR and CMEA a reactor type similar to the PWR is used, referred to as the VVER, cf. Tables 19.1 and 20.1. The older VVER reactors have no containment; except two in Finland (Loovisa 1 & 2) where a Westinghouse type containment was added. The fuel elements have a hexagonal cross section and the steam generators are horizontal rather than vertical. The stainless steel control rods contain boron carbide as neutron absorber. A VVER reactor contains more water in the vessel than a typical PWR. This gives a higher heat capacity allowing a VVER to withstand interrupted coolant flow for a considerably longer time than a PWR. The pressure vessel in the VVER is exposed to rather high neutron fluxes and after many years of operation the wall has to be annealed (heated) in place to remove radiation induced brittleness. The VVER, like the PWR, has to be shutdown for refuelling.

(c) *PHWR*. The advantage of using heavy water for neutron moderation is that its low thermal neutron capture cross section (Σ_a (th), Table 19.3) improves the neutron economy, i.e. $1-p$ becomes much less than in LWRs) so that even natural uranium can be used as fuel. The good neutron economy and harder energy spectrum (cf. Table 19.3) leads to about twice as high plutonium yield in heavy water reactors as compared to LWRs. The first large heavy-water-moderated reactors were built in the United States around 1950 for production of materials for fission weapons. Presently the only commercial PHWR type is the CANDU (CANadian Deuterium Uranium) design. The reactor vessel is a horizontal cylindrical tank filled with D_2O, through which several hundred horizontal aluminum tubes pass. Within each of these *pressure tubes* are bundles of zircaloy clad natural UO_2 pins. The bundles are cooled by pressurized heavy water ($\sim$8 MPa), which in the steam generator produces light water steam for turbines. This pressure-tube-tank design is known as a *calandria*. In some designs the heavy water in the moderator tank can be dumped down into an empty tank below the reactor to allow an immediate shutdown, at the sacrifice of shielding in such an event. In other designs (Bruce, Table 19.7) the calandria is surrounded by a vessel full of ordinary water, which provides shutdown

shielding; shutdown is then achieved by absorber rods and injection of gadolinium nitrate into the moderator. The calandria design permits on-line refuelling.

Early CANDU-PHWR were built in the 200 MW_e size, but the design can be upgraded to 750 MW_e (Table 20.1). Using natural uranium the burnup is comparatively low, about 9 $MW_{th}d\ kg^{-1}$.

The neutron economy makes it feasible to run PHWRs on the thorium cycle with a conversion ratio of ~ 0.9. Such a cycle has to be started and operated either with a mixture of $^{233}UO_2$, enriched $^{235}UO_2$ and ThO_2, or – if isotope enrichment is not used – through the following three step fuel schedule: (i) running on natural uranium and extracting Pu; (ii) running on Th + Pu and extracting ^{233}U; (iii) running on Th + ^{233}U.

The investment cost for CANDU reactors is higher than for LWRs because of a more complicated design and the large amount of D_2O required. The running costs are claimed to be lower; the standard loss of D_2O is $\sim 1\ kg\ d^{-1}$. The neutron capture by D forms tritium in considerable amounts, so the D_2O becomes contaminated by TDO. Also TD and TDO can leak into the surroundings, which may cause difficulties in biological safety. Two methods have been used to limit the amount of T present: i) In an expanding PHWR system part of the D_2O (contaminated by T) in older reactors can be transferred to new reactors and replaced by fresh heavy water thereby providing a temporary relief by dilution, ii) when a troublesome inventory of T has been built-up, the D_2O is transferred to a special isotope separation plant where tritium is removed.

20.1.2. Boiling water reactors

In boiling water reactors (BWR, ATR and RBMK types) the coolant is brought to boiling while passing along the fuel pins. As mentioned in §19.13.2, no separate steam generator and secondary loop exist and the design has less complexity. In addition to negative temperature coefficients both for fuel and moderator, boiling water reactors have a negative *void coefficient*. This means that bubble formation (*void*) along the fuel pins reduces the reactivity. It is therefore common practice to control the reactor power by the main circulation pumps; increased circulation initially reduces the voids (and fuel temperature) thereby increasing power production until the previous void fraction has been almost restored at a slightly higher average fuel temperature. However, a very large negative void coefficient should also be avoided as it can cause an undesirable amplification of pressure fluctuations in the reactor vessel, generating rapidly increasing power oscillations.

In reactors with *pressure suppression* (*PS*) *containments* the steam outlet from the reactor vessel is located in the so-called *upper dry well* (Fig. 19.16). In the case of excess pressure in the tank, safety valves at the steam outlet open, releasing steam into the water in the *wet well* through *blow-down pipes* (the wet well condensation pool typically contains ~ 2000 m^3 water) thereby condensing the steam. Steam from an inadvertent pipe break inside

the containment will also be forced down into the condensation pool. Figure 19.16 shows some emergency systems for the ABB-Atom (Sweden) BWR reactor. In case of multiple failures leading to a too high pressure in the containment, the rupture disk will break and release steam and gas from the wet well through a special filter, thus preventing a failure of the containment.

(a) *BWR*. Most of the typical features of the BWR type is described in §19.13.2. Like the PWR and VVER types, a BWR has to be shut down yearly in order to replace spent fuel. Published data suggests that many BWRs have a marginally higher net efficiency than the typical PWR. The response of a BWR to an increase in power demand is rather slow because of the void effect and it is therefore not used as power plant for propulsion purposes.

(b) *ATR*. The Advanced Thermal Reactor (ATR) was a Japanese D_2O-moderated boiling H_2O cooled MOX-fuelled Pu-burner prototype reactor which is now shut down, see Fig. 20.1. The design has a calandria with vertical pressure tubes, cooled with light water which is allowed to boil in the tubes. This permits on-line refuelling. ATR reactors are seen as a complement to breeder reactors. A large 600 MW_e ATR reactor will require a fuel supply of ∼460 kg Pu per year.

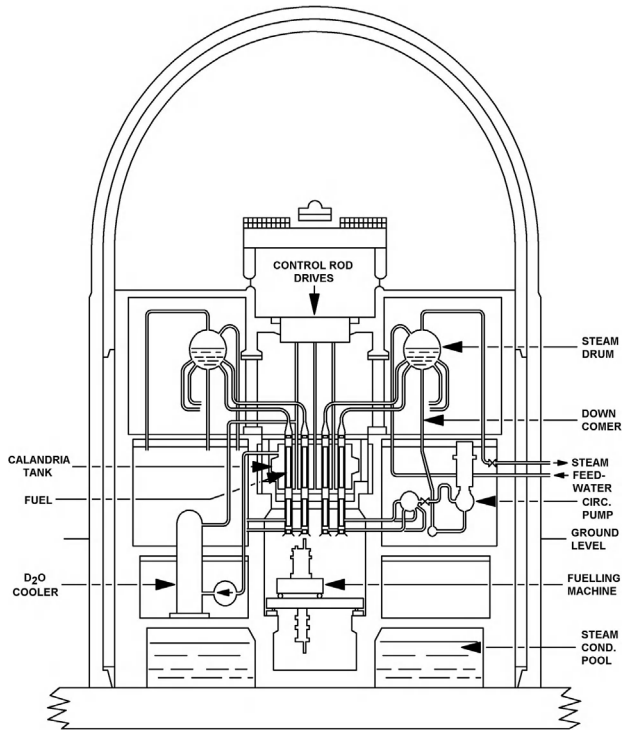


Figure 20.1 Schematic drawing of the Japanese Advanced Thermal Reactor (ATR).

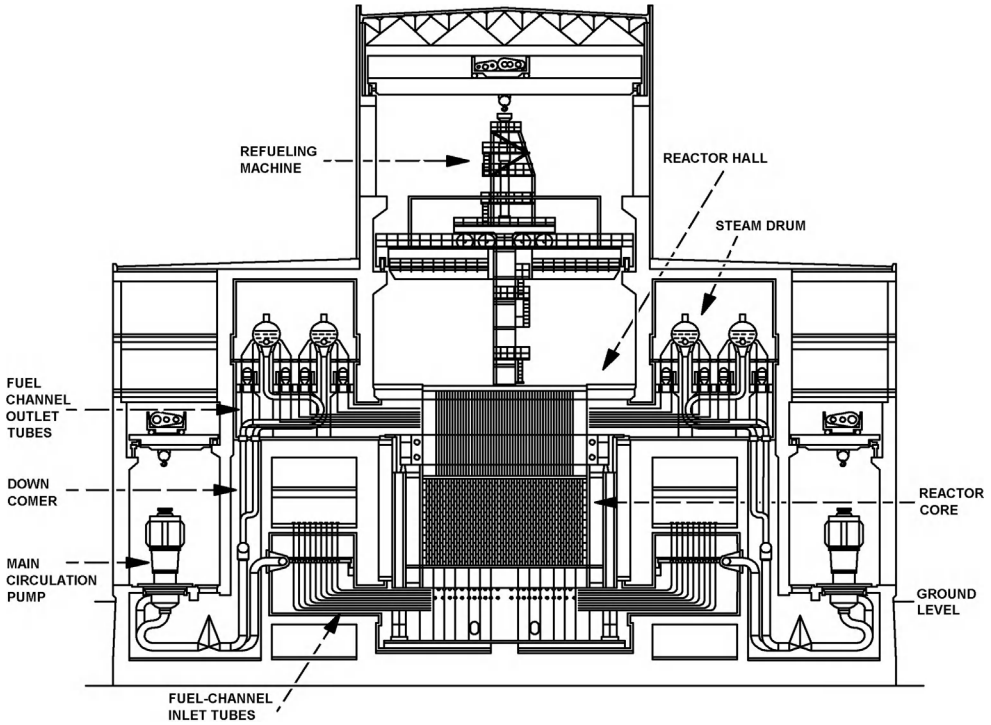


Figure 20.2 Russian graphite-moderated boiling light water cooled power reactor, RBMK.

(c) *RBMK.* The RBMK is a graphite moderated, boiling H_2O cooled, pressure tube reactor of Russian design, see Figure 20.2. The use of many pressure tubes (Zr-Nb alloy), each normally containing two fuel bundles in series, in a large graphite moderator makes it possible to design a boiling water reactor of much larger size and power than possible with a single pressure vessel; up to >2000 MW_e. The typical RBMK has 1000 MW_e output, but two units with 1500 MW_e were built in Lithuania. Furthermore, like in the CANDU design, the use of pressure tubes makes on-line refuelling possible; removing one of the reasons for a yearly shut-down period. Monitoring of individual pressure tubes makes it possible to detect and locate failure rather easily leading to a high plant reliability and availability. Subdivision into several coolant loops also increases the safety of the reactor system. The effective moderator makes it possible to operate the RBMK reactors with fuel only slightly enriched in ^{235}U. Boron carbide containing control rods in special channels through the moderator are used to control neutron flux and power distribution.

The many (1452 in a standard RBMK-1000) pressure tubes makes a pressure tube failure less disastrous than a large pipe break or vessel failure in many other water cooled reactor types. This led to an initial design without any effective containment

(6 RBMK-1000 units). However, the 2nd generation RBMK-1000 (8 units) and the RBMK-1500 (2 units) have a pressure suppression pool for steam condensation in case of pressure tube rupture. Operating statistics shows that the design has been very good with regard to availability and annual load factor.

The disastrous Chernobyl reactor accident highlighted some of the undesirable properties of this reactor type in a dramatic way. The use of a large, very effective, graphite moderator in combination with a light water coolant gives the reactor a large positive void coefficient when operated below $\sim$20% of full power with natural uranium or low enriched fuel; the reactor has a better neutron economy with empty or steam-filled pressure tubes than with water-filled pressure tubes. This feature allowed one of the Chernobyl reactors to increase its power from below 20% to $\sim$100 times full power in about 4 seconds, causing a catastrophe. Furthermore, the large core makes the neutron fluxes in different parts of the core only weakly coupled, e.g. one or more regions of the large core may still be critical when one section is made subcritical by control rods. After the Chernobyl disaster, safety margins in the RBMK reactors have been increased by using higher enrichment in the fuel, special absorbers, an improved control system and better operating procedures. The consequences of this accident are discussed in §13.10.2 and Chapter 22.

It may be noted that the RBMK reactor design was chosen by the USSR (despite warnings from the Soviet Academy of Sciences) because it was better suited to available production facilities than the VVER types (which required the manufacture of large pressure vessels). The rapid introduction of RBMK reactors in the USSR made previously used energy resources (oil and gas) available for export to the West, giving a needed hard currency income.

20.1.3. Special reactor systems

So far we have described some reactors designed for conventional power production. In §19.12, reactors for research or isotope production were discussed. We have purposely not included reactor designs which presently are considered obsolete (e.g. the sodium cooled graphite moderated reactor such as Hallam USA, 75 MW_e), designs which presently seems to have little promise for the future (e.g. gas cooled graphite moderated reactors, GCR), or are unsuited for large scale power production (e.g. the organic moderated and cooled thermal heterogeneous reactor, Piqua, USA, 11 MW_e). Most reactors of these old types have been, or are planned to be, shut down.

Nuclear power reactors may be used for other purposes than large scale production of electricity (or heat) for community consumption. A new type of helium cooled and graphite moderated prototype reactor, HTTR, using a special thorium based high-temperature fuel, has been built at Orai (Japan). The design goal was to heat helium to $\geq$1000°C by the 30 MW_{th} reactor. Presently operation is permitted with a gas outlet temperature of about 950°C. The hot helium produced is intended to drive a new,

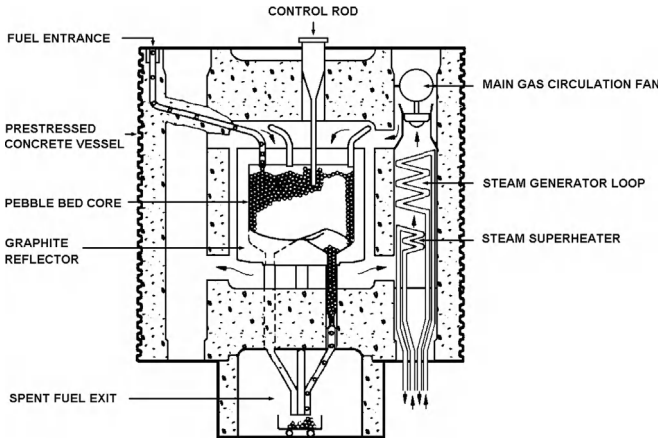


Figure 20.3 Schematic drawing of the German pebble bed reactor (THTR).

continuous, chemical two-cycle process for production of hydrogen from water or from reforming of methane, thus converting nuclear energy into a possible source of hydrogen fuel for combustion engines. A possible alternative is to use the hot helium directly in a He-gas turbine connected to an electrical generator. For safety reasons the HTTR reactor at Orai is located under ground.

A rather unique design is the pebble bed reactor, see Figure 20.3. It is a helium cooled graphite moderated reactor with a core consisting of a bed of spheres; each about 6 cm in diameter. The fuel is initially based on a mixture of $^{235}U/^{232}Th$ carbides as microspheres in a graphite matrix, see §21.7. The fuel spheres are covered with a layer of silica or silicon carbide. This type of reactor can operate as a thermal breeder or converter with its own ^{233}U replacing the ^{235}U in the initial fuel. However, in order to recover new ^{233}U, the spent fuel has to be reprocessed. Spent fuel is withdrawn at the bottom and new fuel added at the top of the core. The possibility to use slightly different sizes of fuel spheres would permit a mechanical sorting of the spent fuel according to initial type before reprocessing. Also this reactor can produce helium at a rather high temperature. A 15 MW_e prototype was operated at Jülich (Germany) from 1967 to 1988 and a larger 300 MW_e reactor was built and operated for about one year (THTR 300). A small prototype HTR-10 is in operation in China.

Both the Japanese and German designs described above have a fuel that can withstand extreme temperatures without melting. Hence a core melt accident in such reactors is practically impossible. Even at high temperatures the fuel integrity is not likely to be compromised, thus strongly reducing the possibility of radioactive releases to the environment in case of a severe accident. However, after air ingress into an overheated core, burning graphite might cause a release of radioactive material. The THTR 300 was permanently shutdown after a minor accident which released some radioactive material into the environment.

The large amount of energy stored in the small volume of fuel makes nuclear reactors suitable as energy sources when repeated refuelling is a disadvantage. For example, a nuclear reactor was used for a long time as a power source in the Arctic. Several nuclear powered merchant ships have been built in order to determine their feasibility, but none is presently in use. The now scrapped USSR icebreaker *Lenin* (44 000 shaft horse power, launched in 1959) had three reactors of the pressurized water type, of which one was a spare. Nuclear power is used in many naval vessels in the USA, Russia, UK, China and France. The first nuclear powered vessel was the US submarine *Nautilus*, launched in 1955; in 1958 it made a famous northern voyage from the Pacific to the Atlantic, passing under the ice at the North Pole. It could travel 250 000 km (~6 times around the earth) without refuelling. At the end of the year 2000, about 250 nuclear powered vessels (8 icebreakers and the rest naval vessels - mostly submarines) were in operation. These vessels correspond to a total of about 400 reactors.

The present nuclear fleets are dominated by advanced submarines which now number several hundred. Their reactors (small PWR:s) are fuelled by enriched uranium metal (20–50% ^{235}U) alloyed with zirconium. The high initial enrichment in combination with burnable poisons and the use of extra neutron shielding to protect the pressure vessel gives a long service life of both the fuel and the reactor; from ~4 years between refuellings for the Russian icebreakers to ~30 years for the 48 MW reactors used in the French Rubis-class submarines. Hence, many submarines today have reactors that can operate without fuel replacement or pressure vessel annealing during the expected life of the submarine itself. Because of their long endurance submerged, high underwater speed, and the difficulty of detecting and destroying these submarines they are considered to act as deterrents against a nuclear war.

The world's first large nuclear powered naval vessel was the US aircraft carrier *Enterprise*, (CVN-65) which has 8 PWR reactors, producing 300 000 shp. With a displacement of ~80 000 t (and ~100 airplanes) it can cruise at 35 knots for 60 days without refuelling.

Finally, 35 small nuclear reactors have been employed by the former USSR and 1 by USA as high power energy sources in satellites. Two naval-type reactors on barges have also been designed in Russia to be used for heat and electricity supply at remote locations in the arctic. Reactors have also been developed and tested in the US for use as aircraft and rocket engines (e.g. NERVA, which was intended as a drive for a manned Mars visit).

20.2. CHEMISTRY OF WATER COOLED REACTORS

The conditions in water cooled reactors are approximately: temperatures $\leq 350°C$, pressures ≤ 16 MPa, and intense γ- and n-irradiation. This causes potentially severe corrosive conditions.